

UBER END-TO-END AZURE DATA ENGINEERING PROJECT

Real-Time Streaming Pipeline with Azure Event Hubs, Databricks & Star Schema

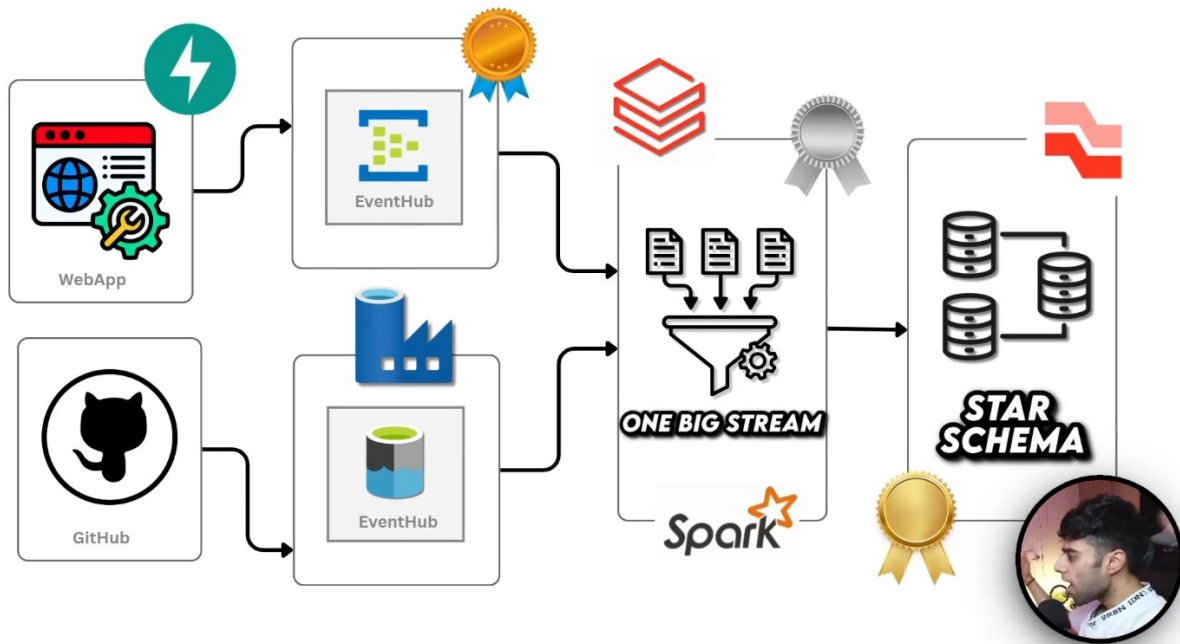


Figure 1: Uber Real-Time Streaming Architecture (WebApp + Event Hubs → Spark → Star Schema)

PROJECT OVERVIEW

This project demonstrates a complete end-to-end real-time data engineering solution for Uber-like ride booking data. It covers the **full pipeline from event generation (WebApp) to real-time streaming processing (Azure Event Hubs + Spark(Spark Declarative Pipeline)) and finally building a production-ready Star Schema with Slowly Changing Dimensions (SCD Type 2) in the Gold layer.**

ARCHITECTURE

The architecture follows a modern real-time streaming pattern:

- Web Application (FastAPI) generates ride booking events
- Azure Event Hubs ingests real-time events (two instances for high throughput)
- Azure Data Factory handles batch/reference data ingestion from GitHub
- Azure Databricks + PySpark Structured Streaming processes the data in real-time
- Spark Declarative Pipelines (SDP) used for bronze → silver → gold transformation
- Final output is a Star Schema model ready for analytics and BI

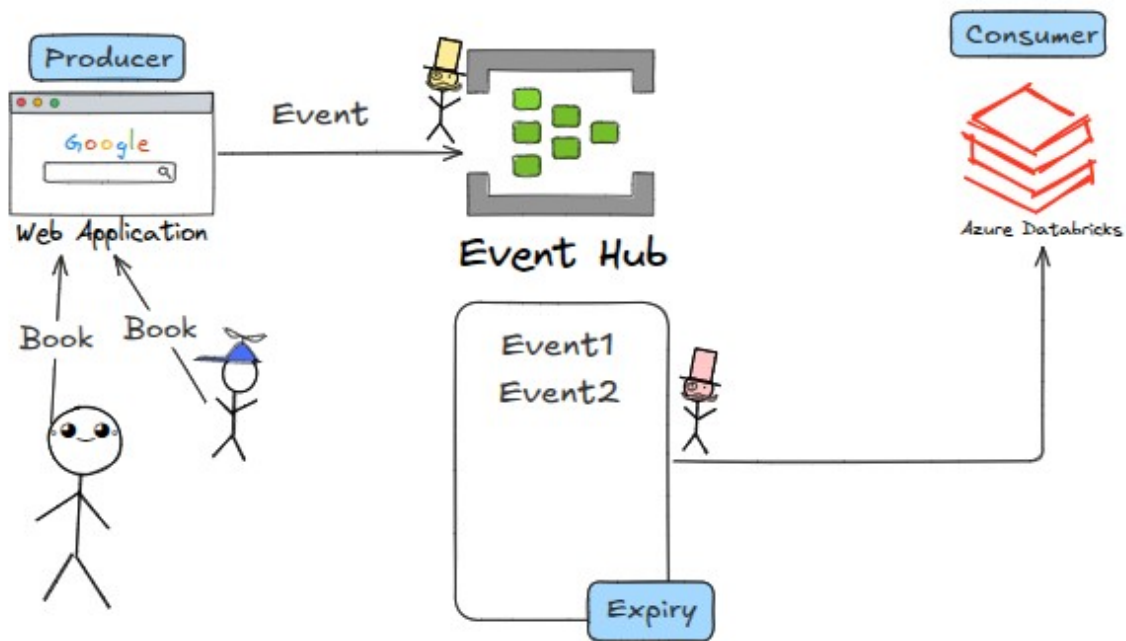
Azure Event Hub

Pub-Sub

Publisher - Subscriber

Pro-Con

Producer - Consumer



can have multiple consumers as well for ex : azure synapse analytics etc.

Events have an expiry duration

[+ Event Hub](#) [Delete](#) [Refresh](#) [Give feedback](#)

Subscription ID : 5342102a-c05b-4637-b821-4ad9e1e6efd6

Host name : uberevents.servicebus.windows.net

Throughput Units : [1 unit](#)

Auto-inflate throughput ... : [Disabled](#)

Local Authentication : [Enabled](#)

Tags ([edit](#)) : [Add tags](#)

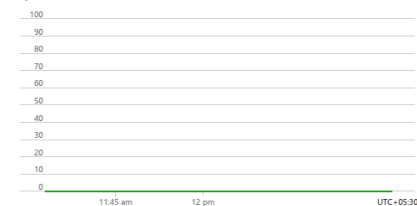
NAMESPACE CONTENTS
0 EVENT HUBS

KAFKA SURFACE
ENABLED

ZONE REDUNDANCY
ENABLED

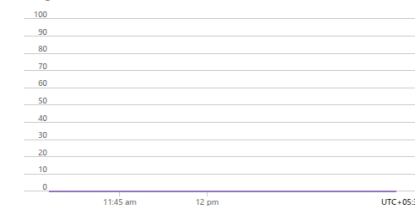
Show data for the last: [1 hour](#) [6 hours](#) [12 hours](#) [1 day](#) [7 days](#) [30 days](#)

Requests



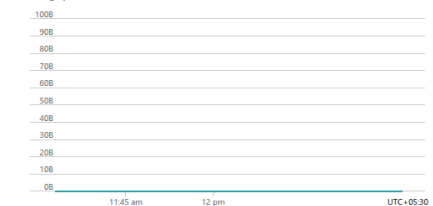
[1/2](#) [Incoming Requests \(Sum\), uberevents](#) | 0
[Successful Requests \(Sum\), uberevents](#) | 0
[Server Errors \(Sum\), uberevents](#) | 0 [User Errors \(Sum\), uberevents](#) | 0

Messages



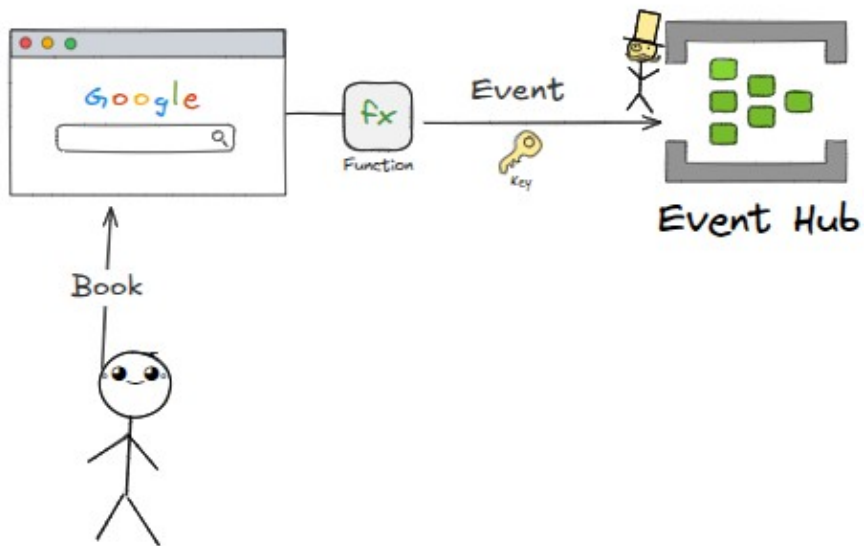
[1/2](#) [Incoming Messages \(Sum\), uberevents](#) | 0
[Outgoing Messages \(Sum\), uberevents](#) | 0
[Captured Messages \(Sum\), uberevents](#) | 0

Throughput

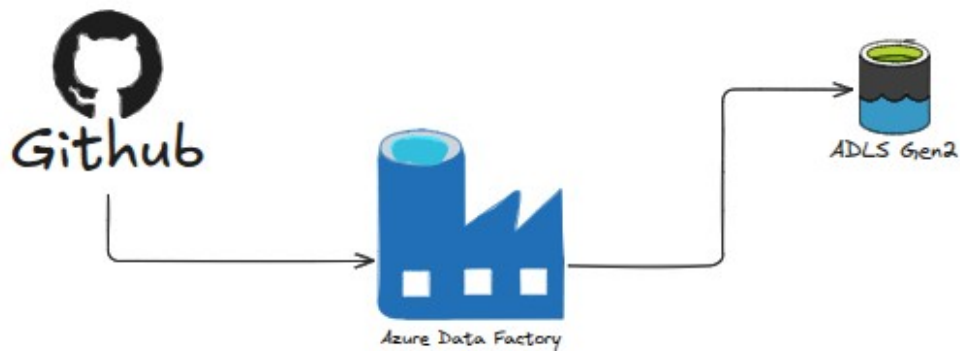


[Incoming Bytes \(Sum\), uberevents](#) | 0B
[Outgoing Bytes \(Sum\), uberevents](#) | 0B
[Captured Bytes \(Sum\), uberevents](#) | 0B

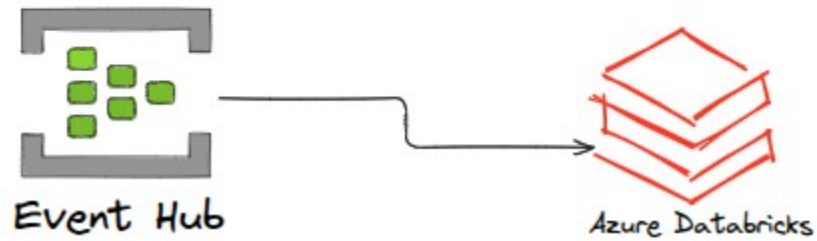
WebApp To EventHub



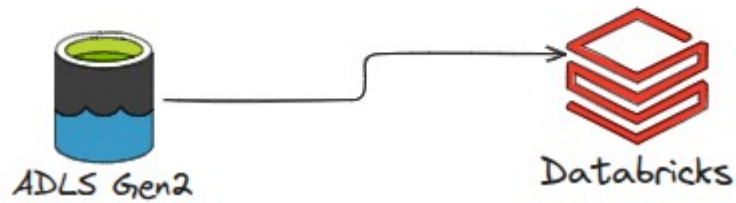
ADF - DataLake



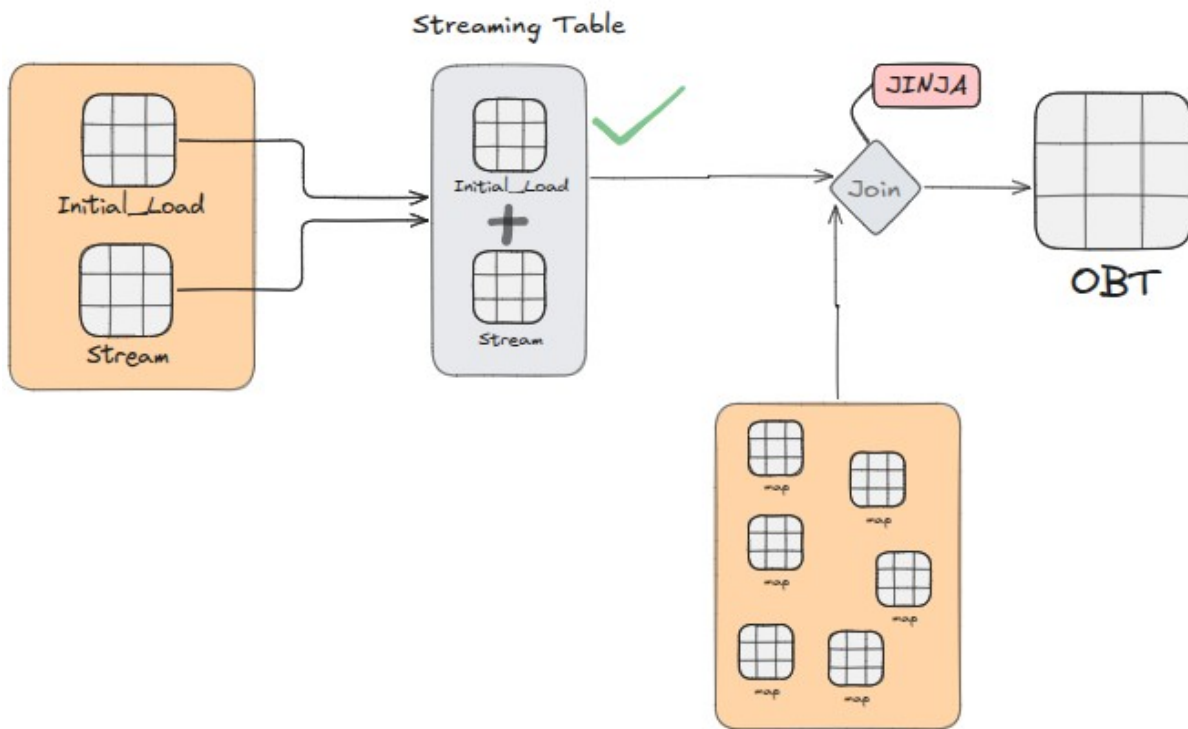
EventHub To Databricks



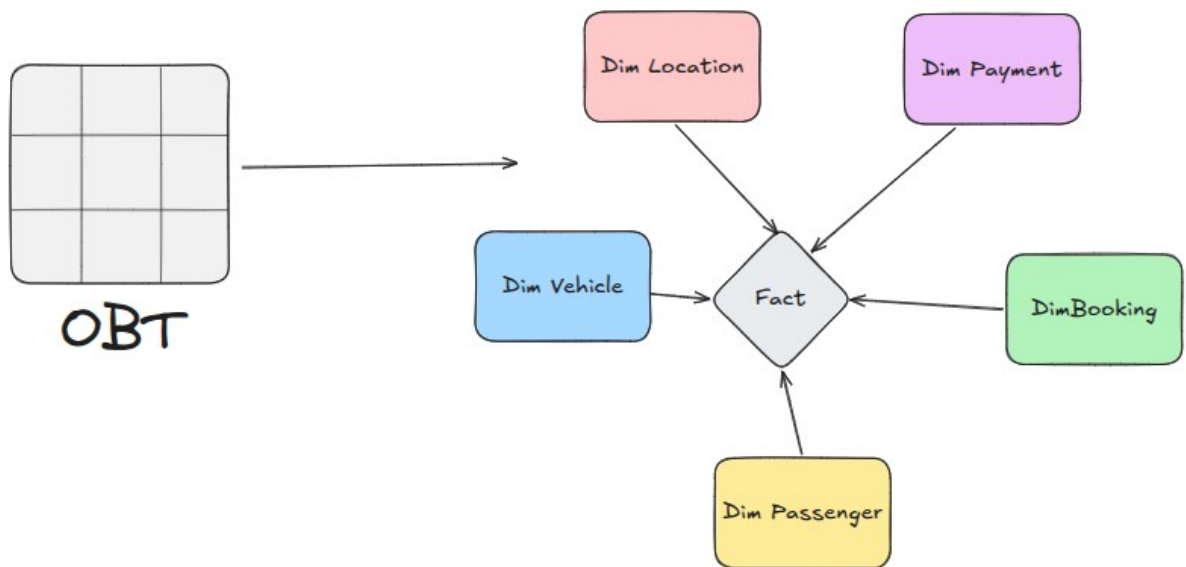
ADLS To Databricks



Silver Layer



Gold Layer



TECHNOLOGY STACK

Layer	Technology	Purpose
Event Ingestion	Azure Event Hubs	Real-time event streaming from WebApp
Batch Ingestion	Azure Data Factory	Reference data & initial load from GitHub
Processing Engine	Azure Databricks + PySpark	Structured Streaming + Declarative Pipelines
Storage	Azure Data Lake Gen2	Bronze, Silver, Gold layers
Data Modeling	Star Schema + SCD Type 2	Fact + Dimension tables with historical tracking
Orchestration	Databricks Workflows	Job scheduling and dependency management
Governance	Unity Catalog	Access control and data lineage

KEY FEATURES

- Real-time event streaming using Azure Event Hubs
- Exactly-once processing with Spark Structured Streaming
- Metadata-driven PySpark transformations
- Declarative pipelines using Spark Declarative Pipelines (SDP)
- Star Schema data model with SCD Type 2 dimensions
- One Big Stream (OBS) / One Big Table (OBT) pattern in Silver layer
- Production-ready CI/CD using Databricks Asset Bundles

IMPLEMENTATION HIGHLIGHTS

- Built a FastAPI WebApp to simulate real-time Uber ride bookings
- Configured Azure Event Hubs with multiple partitions for high throughput
- Implemented PySpark Structured Streaming to read from Event Hubs in real-time
- Created metadata-driven notebooks for Silver layer transformations (joins with mapping tables)
- Built Gold layer with Star Schema (Fact + Dimensions) and SCD Type 2 for location dimension
- Used Spark Declarative Pipelines (SDP) for bronze → silver → gold flow
- Implemented streaming joins with watermarks for late-arriving data

CHALLENGES SOLVED

- Handling high-volume real-time events → Azure Event Hubs with multiple partitions
- Exactly-once processing in streaming → Spark Structured Streaming + checkpointing
- Maintaining historical data in dimensions → SCD Type 2 with start/end dates
- Making transformations maintainable → Metadata-driven Jinja2 templates
- Late-arriving data in streaming joins → Watermark-based streaming joins

SKILLS DEMONSTRATED

- Real-Time Streaming Architecture (Event Hubs + Spark)
- Azure Databricks + PySpark Structured Streaming
- Spark Declarative Pipelines (SDP)

- Star Schema + Slowly Changing Dimensions (SCD Type 2)
- Metadata-Driven Pipeline Development
- One Big Stream / One Big Table (OBS/OBT) Pattern
- Streaming Joins with Watermarks
- Production CI/CD with Databricks Asset Bundles